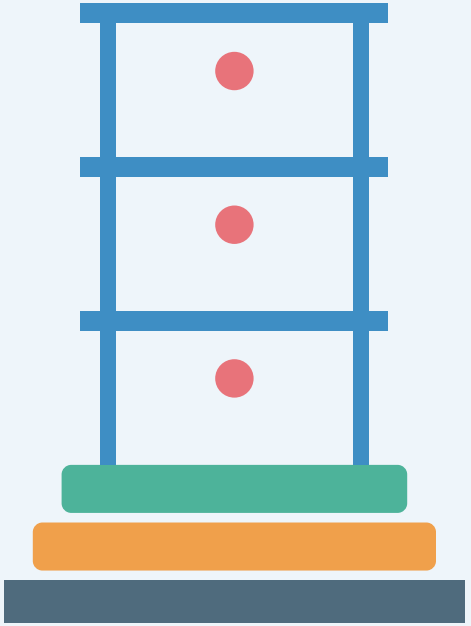
PROJECT PROPOSAL · CENG 396

STRATA

Active Base Stabilization and Real-Time Structural Digital Twin



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Instructor: Erkan Zergeroğlu

We build safety, not assumptions.

Our Team



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The Problem

Horizontal ground motion can excite structural vibration and amplify floor sway.

A useful experiment must measure both the disturbance at the base and the actual motion of the building — instead of assuming that a level-looking platform is stable.

Measure, don't assume.



Base disturbance

What enters the system?



Building response

How much do the floors sway?

Our Solution: How It Works

1. Shake

The lower base is disturbed by hand along one horizontal (X) axis.

2. Measure

Position feedback tracks the base vs. the table and the stage vs. the base.

3. Counter-move

The local controller drives the motor to move the upper stage against the base motion.

4. Observe

Camera and IMUs record the input and response; the dashboard shows a live digital twin.

System Architecture: Four Layers

04 Flexible building & markers — Three-story model with visual markers and IMUs

03 Motor-driven upper stage — Produces controlled counter-motion

02 Hand-shaken lower base — Manual disturbance along the X axis

01 Fixed reference table — Stationary reference for all measurements

A fixed external camera observes all layers. Motion is limited to left–right translation.

Four Independently Testable Modules

MODULE 1 · ≥ 4 students

Physical Stabilization & Control

Bench-tested, safely moving active stage

MODULE 2 · ≥ 2 students

Inertial Sensing & Data

Reliable timestamped IMU data stream

MODULE 3 · ≥ 2 students

Computer Vision & Tracking

Live marker displacement in millimeters

MODULE 4 · ≥ 2 students

Digital Twin & Analytics

Dashboard tested with mock or real streams

Module 1: Physical Stabilization & Control

≥ 4 students

Builds both linear motion layers and the building model; selects the actuator; implements a safe closed-loop controller with stroke limits and emergency stop.

Input

Lower-base position vs. table; upper-stage position vs. base; mode and safety commands

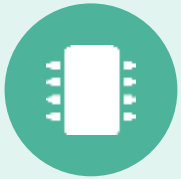
Output

Motor command, stage displacement, limit/fault flags, timestamped control status

Independent test

Scripted position requests on the bench, then shake the base; test physical stops and loss-of-signal before mounting the building

Module 2: Inertial Sensing & Data



≥ 2 students

Provides calibrated motion measurements from the base and the structure, independent of the camera and the software.

Input

Raw IMU acceleration / angular-rate samples, device settings, time reference

Output

Timestamped calibrated acceleration, sensor health flags, replayable recordings

Independent test

IMU on a hand-shaken test board: check static gravity, orientation, noise, repeatability; replay logs for filtering

Module 3: Computer Vision & Tracking



≥ 2 students

Uses a fixed external camera and visual markers to measure horizontal displacement of the base and each floor.

Input

Camera frames, reference markers, calibration scale, timestamps

Output

Base and floor X displacement (mm), tracking confidence, marker-loss events

Independent test

Move a marked board by known distances (10 mm, 20 mm) and compare with a ruler; test blur and occlusion on recorded video

Module 4: Digital Twin & Analytics

≥ 2 students

Displays a live structural model with synchronized plots; compares measurements with simple model predictions without mixing the two.

Input

Actuator status, IMU signals, marker positions (mock or replayed during development)

Output

Live visualization, time-series charts, peak/RMS metrics, ON/OFF comparison, trial reports

Independent test

Stream synthetic waveforms and mock JSON logs; validate plotted values, export, and missing-data handling

All Modules Together

PHYSICAL SYSTEM · table · hand-shaken base · motor stage · building with markers

Module 1

Physical Stabilization & Control

Actuator status, stage displacement, fault flags

Module 2

Inertial Sensing & Data

Calibrated acceleration, health flags

Module 3

Computer Vision & Tracking

Base & floor X (mm), confidence

SHARED DATA CONTRACT

Module 4 · Digital Twin & Analytics

Live model, synchronized plots, ON/OFF comparison

Shared Data Contract

Direction
Signed X direction (single horizontal axis)

Units
mm for displacement, m/s² for acceleration

Metadata
Sensor source and timestamps

Quality
Confidence / health flags

Control state
Actuator limits and ON/OFF state

Development
Non-critical modules can start with generated or replayed data

Final Demo: Control OFF vs. Control ON

A — Control OFF

What happens

Upper stage held at a fixed reference position relative to the lower base.

What we measure

Base input, roof displacement and acceleration

B — Control ON

What happens

Feedback controller drives the upper stage to oppose the measured base motion.

What we measure

Same responses, plus actuator travel and control signal

Same structure · same manual input · comparable disturbance

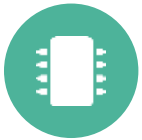
Measurement and Success Criteria

Three independent sources observe every trial



Fixed camera

Tracks markers on the base and each floor



IMUs

Measure acceleration



Encoders / linear sensors

Stage-position feedback



Success measure

- Compare top-floor peak and RMS displacement and acceleration (OFF vs. ON)
- Record input disturbance, actuator travel and controller limits
- Show plots with a synchronized live visualization

Scope and Expectations

What it is

- Low-amplitude, one-direction demonstrator
- Educational and measurable
- Outcomes are targets to be tested, not guaranteed reduction figures

What it is not

- A real-earthquake simulator
- A building safety certification
- Protection from all disturbances

Thank You

Questions?



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